

4 (a)	From infinity to Mars, $\text{KE gain} = \text{GPE loss}$ Thus, $\frac{1}{2}mv^2 - 0 = m_{\text{surface}} - m$ $= (-\frac{GMm}{R}) - \text{zero}$ □ speed of rock $v = \frac{2GM}{D}$ where $M = 6.4 \times 10^{23} \text{ kg}$, $D = 6.8 \times 10^6 \text{ m}$ $= 5.0 \times 10^3 \text{ m s}^{-1}$	[1] [1]
(b)(i)	$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$ □ temperature $T = \frac{m\langle c^2 \rangle}{3k}$ where mass of 1 atom, $m = \frac{4 \times 10^{-3} \text{ kg}}{N_A}$, $\langle c^2 \rangle = [\text{Ans in (a)}]^2 = 2.5 \times 10^7 \text{ m}^2 \text{ s}^{-2}$ $= 4.0 \times 10^3 \text{ K}$ • [1 m] for mass of 1 atom, $m = \frac{4 \times 10^{-3} \text{ kg}}{N_A}$ • [1 m, ecf v & m] for correct caln of T	[1] [1]
(ii)	1. The speed in (a) is the escape speed fr surface of Mars. 2. Since surface temperature of Mars is (much) lower than the temperature in (b) (i), 3. rms speed of He-4 is (much) lower than the escape speed. Thus, helium-4 gas is found on surface of Mars.	[1] [1]

5(a)	Progressive: all particles have the same amplitude	
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